

torch.quantize_per_tensor#

https://pytorch.org/docs/stable/generated/torch.quantize_per_tensor.html

Description:

Converts a float tensor to a quantized tensor with given scale and zero point. input (Tensor) – float tensor or list of tensors to quantize scale (float or Tensor) – scale to apply in quantization formula zero_point (int or Tensor) – offset in integer value that maps to float zero dtype (torch.dtype) – the desired data type of returned tensor. Has to be one of the quantized dtypes: torch.qint8, torch.qint8, torch.qint32

Function Signature

`torch.quantize_per_tensor(input, scale, zero_point, dtype) → Tensor#`

Parameters

Returns

Return type

Returns:

Tensor

Code Examples

```
>>> torch.quantize_per_tensor(torch.tensor([-1.0, 0.0, 1.0, 2.0]), 0.1, 10, torch.quint8)
tensor([-1., 0., 1., 2.], size=(4,), dtype=torch.quint8,
      quantization_scheme=torch.per_tensor_affine, scale=0.1,
      zero_point=10)
>>> torch.quantize_per_tensor(torch.tensor([-1.0, 0.0, 1.0, 2.0]), 0.1, 10, torch.quint8).int_repr()
tensor([ 0, 10, 20, 30], dtype=torch.uint8)
>>> torch.quantize_per_tensor([torch.tensor([-1.0, 0.0]),
torch.tensor([-2.0, 2.0])],
>>> torch.tensor([0.1, 0.2]), torch.tensor([10, 20]),
torch.quint8)
(tensor([-1., 0.], size=(2,), dtype=torch.quint8,
      quantization_scheme=torch.per_tensor_affine, scale=0.1,
      zero_point=10),
 tensor([-2., 2.], size=(2,), dtype=torch.quint8,
      quantization_scheme=torch.per_tensor_affine, scale=0.2,
      zero_point=20))
>>> torch.quantize_per_tensor(torch.tensor([-1.0, 0.0, 1.0, 2.0]), torch.tensor(0.1), torch.tensor(10), torch.quint8)
tensor([-1., 0., 1., 2.], size=(4,), dtype=torch.quint8,
      quantization_scheme=torch.per_tensor_affine, scale=0.10,
      zero_point=10)
```

Detailed Documentation

Documentation

Converts a float tensor to a quantized tensor with given scale and zero point.

input (Tensor) – float tensor or list of tensors to quantize

scale (float or Tensor) – scale to apply in quantization formula

zero_point (int or Tensor) – offset in integer value that maps to float zero

dtype (torch.dtype) – the desired data type of returned tensor. Has to be one of the quantized dtypes: torch.qint8, torch.qint8, torch.qint32

A newly quantized tensor or list of quantized tensors.